

Omar Ali Alkhayat

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Summary

Results-driven Backend Software Engineer with extensive hands-on expertise in architecting scalable backend systems, robust REST APIs, and advanced AI integrations. Proficient in Python/Django alongside practical knowledge of modern frameworks like FastAPI, Spring Boot, and NestJS. Proven leadership capabilities in guiding cross-functional development teams through complex, high-impact technical projects from conception to production. Passionate about clean architecture, performance optimization, and solving intricate engineering challenges

Technical Skills

- **Backend Frameworks:** Django, Django REST Framework, FastAPI, Spring Boot, NestJS
- **Programming Languages:** Python, C++, HTML, JavaScript, TypeScript
- **AI & Machine Learning:** Neural Networks, Deep Reinforcement Learning (PPO, IPPO/MAPPO), RAG pipelines, LLM integration, PyTorch, scikit-learn
- **Databases & Tools:** MySQL, SQL Server, SQLite, Elasticsearch, Git, Linux, REST APIs

Software Engineering Experience

Multi-Agent Reinforcement Learning for Adaptive Traffic Signal Control 2026

Backend & AI Developer | Graduation Project (Team of 2)

- Engineered a reinforcement-learning traffic controller coordinating nine signalized junctions within a SUMO simulation.
- Implemented IPPO and MAPPO algorithms from scratch, utilizing a continuous green-duration policy.
- Designed a multi-term reward system optimizing queues, delay, throughput, and spillback, benchmarking across 100 paired seeds.
- Successfully fine-tuned the final model onto real-world road networks in Damascus.
- **Tools:** Python, PyTorch, SUMO/TraCI, Petting Zoo, Gymnasium, NumPy, pandas

Document Search Engine 2024

Backend Engineer | team of 3

- Architected a modular Python AI Chat Assistant featuring dual Rich terminal and Tkinter GUI interfaces, multi-LLM routing, persistent session storage, and intelligent rate-limiting with retry mechanisms.
- Developed the core LLM abstraction layer and delivered a polished, production-ready frontend and backend integration.
- **Tools:** Python, Rich, Tkinter, LLM APIs

Banking Management System

2024

Technical Team Lead & Full Stack Engineer | team of 3

- Spearheaded the development of a comprehensive banking system managing employee, customer, loan, and branch operations.
- Architected, coded, and deployed the Branch and Employee Management modules from the ground up.
- Led a team of developers, ensuring clean architecture, robust code quality, and timely project delivery.

Real-Time Fleet Management Platform

2025

Backend Engineer (Solo Project)

- Developed a platform for continuous vehicle telemetry and geospatial monitoring.
- Solely engineered the entire backend infrastructure, implementing low-latency data processing and scalable API architecture.
- **Tools:** Python, Django REST Framework, REST APIs

HIV Therapy Optimization with Genetic Algorithms

2025

AI Backend Engineer | team of 2

- Engineered an AI-driven clinical decision support system optimizing HIV antiretroviral therapy protocols based on WHO 2024 guidelines.
- Architected a multi-objective fitness engine balancing clinical efficacy, toxicity, cost, and patient adherence.
- **Tools:** Python, Genetic Algorithms, Stanford HIVDB

AI Chat Assistant

2025

AI Backend Engineer (Solo Project)

- Architected a modular Python AI Chat Assistant featuring dual Rich terminal and Tkinter GUI interfaces, multi-LLM routing, persistent session storage, and intelligent rate-limiting with retry mechanisms.
- Developed the core LLM abstraction layer and delivered a polished, production-ready frontend and backend integration.
- **Tools:** Python, Rich, Tkinter, LLM APIs

MTNSYR - Telecom Data Warehouse & Analytics Platform

2026

Data & Backend Engineer | Team of 2

- Built an ETL pipeline reconciling telecom subscriber data into a robust five-table star schema.
- Developed a Random Forest customer-lifetime-value model and spend-segment classifier.
- Designed and deployed a comprehensive 9-page analytics dashboard accessible via a single CLI.
- **Tools:** Python, pandas, scikit-learn, SQLite, Streamlit, Plotly

Education

B.Sc. in Computer Engineering and Informatics

2021 – Aug 2026

Antioch Syrian University

Certifications & Courses

- **IBM:** Software Engineering, Django Application Development with SQL, Python for Data Science & AI, Developing AI Applications (Flask), Linux Commands, Git Version Control.
- **University of Michigan:** Web Application Technologies and Django.

Languages

Arabic (Native) | English (Professional Working Proficiency)

